

ODLC System Implementation

1. Introduction :

This report documents the design, implementation, and evaluation of an Object Detection, Localization, and Classification (ODLC) pipeline for aerial imagery. The system detects and classifies standard objects — colored uppercase alphanumeric characters placed on colored geometric shapes — and localizes them in real-world coordinates. The work covers dataset construction, model selection and training, evaluation, and localization.

2. Dataset Preparation :

2.1 Data Source and Construction :

A synthetic dataset was generated:

- Backgrounds: Real top-down view runway imagery collected from Kaggle.
- Objects: Synthesized and overlaid using a custom Python script built on the PIL (Pillow) library. Each object consists of:
 - An alphanumeric character (A–Z, 0–9), uppercase
 - Placed on a geometric shape (circle, semicircle, quarter circle, triangle, rectangle, pentagon, star, cross)
 - Using one of the 8 required colors (white, black, red, blue, green, purple, brown, orange)
- Total images: 3,000
- Split: Training 2,000 / Validation 500 / Testing 500 (70/15/15)

2.2 Sample Images :



2.3 Augmentation :

Two offline augmentations were planned: Gaussian blur (applied post-rendering) and random rotation. However:

- Blur was removed after it caused instability and slowdown.
- Rotation was inadvertently omitted during the final dataset export.

Training therefore proceeded on the raw images. Generalization remained thanks to:

- High intrinsic variability in the synthetic data (8 colors, 8 shapes, 36 characters, varied scales and backgrounds).
- On-the-fly augmentation performed by Ultralytics YOLO during training — including *mosaic* (disabled in the last 10 epochs), *HSV jitter*, *random scale/translation*, *horizontal flip*, and *random erasing*.

Lesson learned: Offline augmentation should be validated on a small batch before full training, and the dataset should be version-controlled.

3. Model Selection and Training :

3.1 Model Choice: YOLO26n (Ultralytics) :

We selected YOLO26n, initialized from COCO-pretrained weights (yolo26n.pt).

Justification:

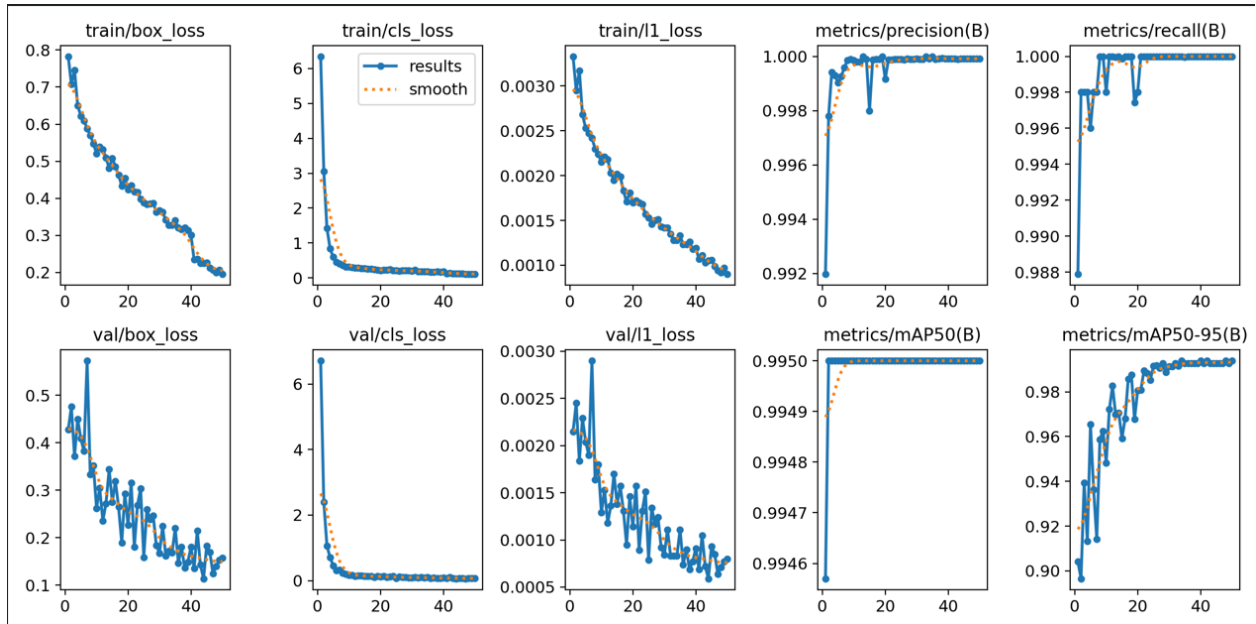
1. State-of-the-art single-stage detector — improved backbone (C3k2 blocks, C2PSA attention), SPPF, and a decoupled head, offering an excellent accuracy-to-latency trade-off.
2. Nano scale (~ 2.5 M params, 5.9 GFLOPs) — enabling real-time inference suitable for UAV on-board deployment.
3. Transfer learning from COCO — faster convergence on the modest 3,000-image dataset ("*Transferred 606/708 items*").
4. Mature framework — built-in augmentation, AMP, reproducible logging.
5. Single-class formulation (target) — defers fine-grained classification downstream and maximizes recall.

3.2 Training Configuration:

Parameter	Value
Model	yolo26n.pt (COCO-pretrained)
Epochs	50
Image size	640 × 640
Batch size	16
Device	NVIDIA Tesla T4
Optimizer	AdamW (auto-selected), $lr_o = 0.002$
AMP	Enabled
Mosaic	1.0 (disabled last 10 epochs)
Training time	≈ 48 minutes (0.80 h)

4. Training and Validation Metrics:

4.1 Training Curves:



Interpretation:

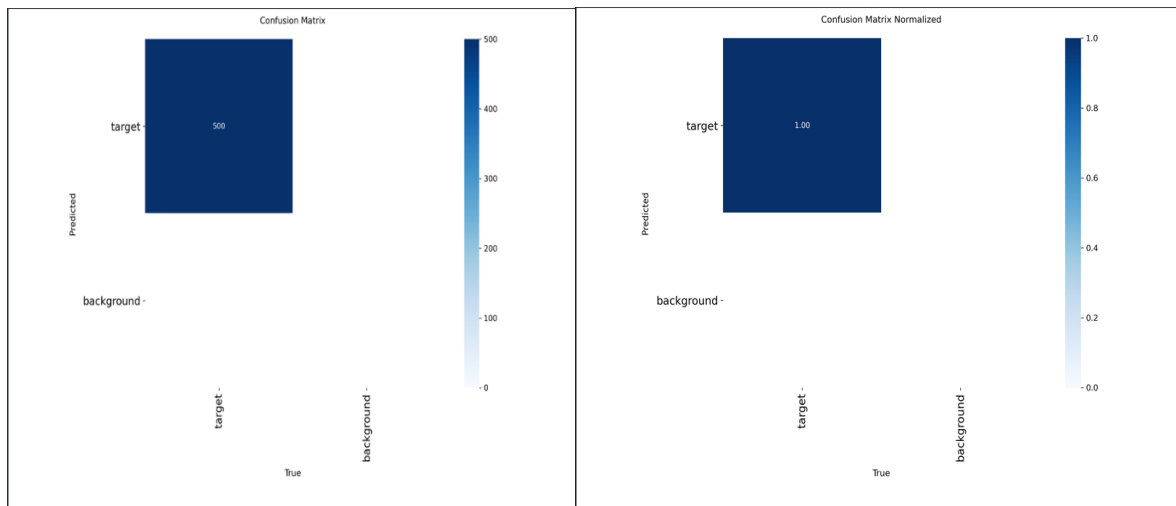
- Train losses (box_loss, cls_loss, l1_loss) decrease smoothly across all 50 epochs — stable optimization without overfitting.
- Validation losses track training losses — good generalization.
- mAP@0.5 saturates at 0.995 from epoch 1.
- mAP@0.5:0.95 climbs steadily from 0.904 → 0.994.

4.2 Final Metrics ([best.pt](#), 500-image validation set, 50 epochs) :

Metric	Value
Precision (P)	1.000
Recall (R)	1.000
mAP@0.5	0.995
mAP@0.5:0.95	0.994

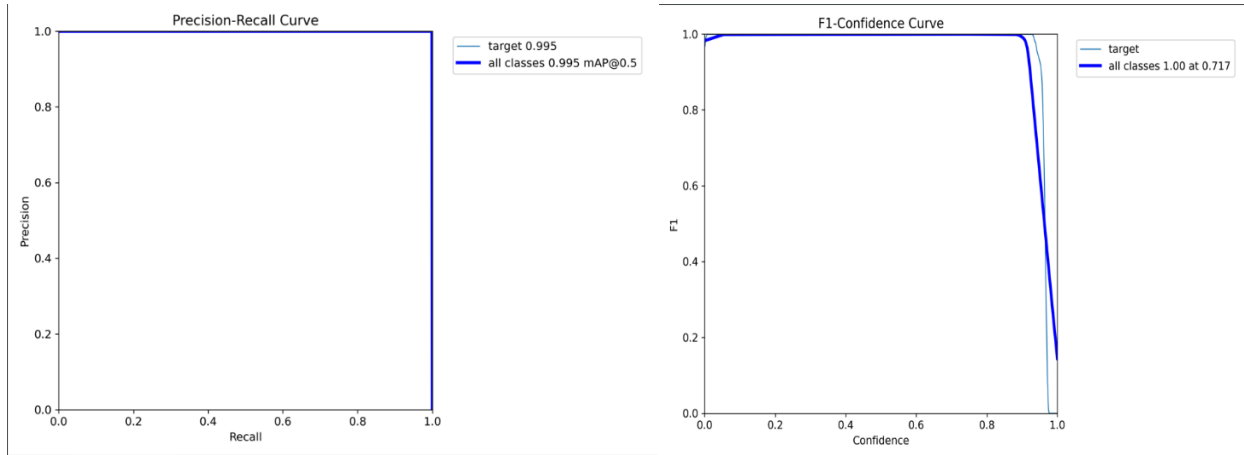
4.3 Confusion Matrix:

The confusion matrix shows 500/500 correct predictions for the single target class with zero false positives and zero false negatives.



4.4 PR / F1 Curves :

The PR curve reaches AUC ≈ 0.995 ; the F1–Confidence curve peaks at F1 = 1.00 at confidence ≈ 0.717 , wide operating range.



5. Object Localization:

After detection, the bounding box center (c_x, c_y) is projected onto the ground plane using the SIYI A8 Mini pinhole camera model, under the task assumptions: nadir-looking camera, altitude $h = 80$ m, focal length $f = 21$ mm, and image center as origin.

5.1 Pixel-to-Meter Conversion :

The Ground Sample Distance (GSD) — the physical distance per pixel — is computed per axis as:

$$\text{GSD}_x = (h \times \text{sensor_width}) / (f \times \text{image_width})$$

$$\text{GSD}_y = (h \times \text{sensor_height}) / (f \times \text{image_height})$$

with sensor = 7.6 mm $\times$ 5.7 mm for the SIYI A8 Mini.

5.2 Ground Coordinates :

The pixel offset from the image center is:

$$\Delta x_{px} = c_x - w/2$$

$$\Delta y_{px} = h/2 - c_y \quad (\text{sign inverted: image y grows downward})$$

and the metric coordinates are:

$$X_m = \Delta x_{px} \times \text{GSD}_x$$

$$Y_m = \Delta y_{px} \times \text{GSD}_y$$

This places the object on a local East-North plane centered at the principal point.

5.3 Integration and Sample Result :

The localization module is integrated into the inference pipeline: for each detection, it extracts the bounding box, computes the GSD, converts the pixel center to (X, Y), and renders the object center as a red dot, the box in blue, and the coordinates as a yellow label.



The figure above shows a green cross labeled "Y" detected with confidence 0.96. Its bounding-box center at pixel (361.8, 206.5) corresponds to:

- X = +1.89 m (East of image center)
- Y = +3.85 m (North of image center)

confirming that pixel-space detections are correctly mapped to metric ground coordinates under the task's assumptions.

6. Discussion:

Strengths:

- Near-perfect detection metrics ($\text{mAP@0.5} = 0.995$).
- Real-time inference (~ 50 FPS on a Tesla T4).
- Robust detection across diverse backgrounds.

Limitations:

- Metrics were computed on a synthetic validation set, a real-image spot check succeeded.
- The single-class formulation defers shape/color/character classification down streams.

7. Conclusion:

A complete ODLC detection pipeline was built using a YOLO26n detector trained on a 3,000-image synthetic dataset. The model achieved $\text{mAP@0.5} = 0.995$ and $\text{mAP@0.5:0.95} = 0.994$ with precision = recall = 1.000, trained in under one hour on a single Tesla T4. A pixel-to-metric localization module was integrated and produced accurate ground coordinates (e.g., $X = +1.89$ m, $Y = +3.85$ m). The results demonstrate that synthetic data — combined with COCO transfer learning and YOLO's built-in on-the-fly augmentation — is sufficient for a high-performance ODLC system.